MATH 284 MATLAB Project #1 - Linear System & Matrix Transformation Explorer

$$
\begin{aligned}
-x+3y&=-72\\
3x+4y-4z&=-4\\
-20x-12y+5z&=-50
\end{aligned}
$$

$$
\begin{aligned}
16x+16y+17z&=10\\
-14x+17y-3z&=75\\
-5x-11y-18z&=43
\end{aligned}
$$

Project Objective
Create a MATLAB GUI application that allows users to enter and analyze linear systems, visualize solutions in linear systems of equations, perform matrix operations, and explore how matrices represent transformations.

The application acts as an interactive introduction to linear algebra concepts from Chapter 1 of the textbook.

Concepts And Textbook Chapters/Sections Covered
Chapter 1 - Linear Equations in Linear Algebra

- 1.1 Systems of Linear Equations
- 1.2 Row Reduction and Echelon Forms
- 1.3 Vector Equations
- 1.4 Matrix Equation ($Ax = b$)
- 1.5 Solution Sets of Linear Systems
- 1.6 Applications of Linear Systems
- 1.7 Linear Independence
- 1.8 Intro to Linear Transformations
- 1.9 Matrix of a Linear Transformation

- Systems of linear equations
- Row reduction
- Echelon forms
- Vector equations
- Matrix equations $Ax=b$
- Solution sets
- Linear independence
- Linear transformations
- Transformation matrices

GUI Layout

```
-----------------------------------------------------
| Linear Algebra Explorer
-----------------------------------------------------
| Input System
|
| Matrix A:               Vector b:                 |
| [             ]         [        ]
|
| [Analyze System] [Row Reduce] [Clear]
-----------------------------------------------------
| Results Panel
|
| RREF Matrix:
|
| Solution Type:
|
| Solution Vector:
-----------------------------------------------------
| Visualization
|
|     2D Vector Plot / Transformation Plot
|

-----------------------------------------------------
| Tools
| [Vector Check] [Independence Test]
| [Transformation Demo]
-----------------------------------------------------
```

Necessary MATLAB Files For This Project

LinearAlgebraExplorerApp.m

Main GUI File.

Responsibilities:

- ***Create GUI***
- ***Manage user inputs***
- ***Connect buttons to functions***
- ***Display outputs***

LinearSystem.m

This file represents the matrix equation $Ax = b$.

VectorAnalyzer.m

This file covers:

- ***Vector equations***
- ***Linear combinations***
- ***Independence***

LinearTransformation.m

This file represents the equation $T(x) = Ax$.

LinearPlotter.m

This file handles visualization.

Necessary MATLAB Methods For This Project

createComponents()

analyzeSystem()

performRowReduction()

solveSystem()

displayResults()

clearGUI()

rowReduce()

solve()

determineSolutionType()
getRank()
checkIndependence()
spanCheck()
linearCombination()
applyTransformation()
plotTransformation()
plotVectors()
plotTransformation()
plotSolution()

GUI Functions

Enter and solve systems
Show REF/RREF
Linear combination calculator
Matrix solver
Determine infinite/no/unique solutions
Small modeling examples
Independence checker
Apply transformations
Display transformation matrix

Pseudocode

START PROGRAM

Create GUI window

Initialize:
    Matrix input fields
    Vector input fields
    Result display
    Graph area

WHEN Analyze Button pressed:

Read A matrix
Read b vector

Create LinearSystem object

Calculate:
RREF
rank
solution type

Display results

WHEN Row Reduce pressed:

Perform Gaussian elimination

Display REF and RREF

WHEN Independence pressed:

Get user vectors

Create VectorAnalyzer object

Determine:
Independent
Dependent

WHEN Transformation pressed:

Get transformation matrix

Get input vector

Calculate Ax

Plot original and transformed vector

WHEN Clear pressed:

Reset all fields

END PROGRAM